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Research Article

Utilizing Age-Adaptive Deep Learning Approaches for Detecting Inappropriate Video Content

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The exponential growth of video-sharing platforms, exemplified by platforms like YouTube and Netflix, has made videos available to everyone with minimal restrictions. This proliferation, while offering a variety of content, at the same time introduces challenges, such as the increased vulnerability of children and adolescents to potentially harmful material, notably explicit content. Despite the efforts in developing content moderation tools, a research gap still exists in creating comprehensive solutions capable of reliably estimating users' ages and accurately classifying numerous forms of inappropriate video content. This study is aimed at bridging this gap by introducing VideoTransformer, which combines the power of two existing models: AgeNet and MobileNetV2. To evaluate the effectiveness of the proposed approach, this study utilized a manually annotated video dataset collected from YouTube, covering multiple categories, including *safe*, *real violence*, *drugs*, *nudity*, *simulated violence*, *kissing*, *pornography*, and *terrorism*. In contrast to existing models, the proposed VideoTransformer model demonstrates significant performance improvements, as evidenced by two distinct accuracy evaluations. It achieves an impressive accuracy rate of (96.89%) in a 5-fold cross-validation setup, outperforming NasNet (92.6%), EfficientNet-B7 (87.87%), GoogLeNet (85.1%), and VGG-19 (92.83%). Furthermore, in a single run, it maintains a consistent accuracy rate of 90%. Additionally, the proposed model attains an $F1$-score of 90.34%, indicating a well-balanced trade-off between precision and recall. These findings highlight the potential of the proposed approach in advancing content moderation and enhancing user safety on video-sharing platforms. We envision deploying the proposed methodology in real-time video streaming to effectively mitigate the spread of inappropriate content, thereby raising online safety standards.

Keywords: age-based ranking; inappropriate content; video classification

1. Introduction

Video-sharing platforms, such as YouTube and Netflix, have ascended to unprecedented levels of popularity. YouTube has grown exponentially and become one of the largest websites for entertainment, learning, and news/information broadcasting. A study published in 2023 [1] has argued that over 2 billion people watch YouTube videos of diverse interests every month. People spend a lot of their leisure time on YouTube, and it is estimated that they spend about 40 min (average) per visit. There is always something new to watch on YouTube because more than 500 h of videos are added every minute. It is also reported that people watch over 1 billion hours of videos on YouTube every day, having diverse watching interests ranging from entertainment, educational videos, sports, and news. Most of these views come from smartphones, tablets, and other pervasive devices. This makes up more than 70% of the watching contribution compared to other devices. YouTube also made money from advertisements (ads), and over 15 billion dollars in revenue was created in 2023. This makes it a great platform for businesses, education, news channels, individuals sharing videos, and advertisers to reach lots of people [2]. YouTube's largest demographic comprises individuals aged 18–34, constituting approximately 38% of its entire user base.

Moreover, YouTube's global appeal is undeniable, as it is accessible in over 100 countries and supports 80 different languages, which makes it a truly worldwide platform for sharing and watching content.

Watching videos on such social media sites, for example, YouTube, can have a significant positive impact on children, contributing to their cognitive, social, and emotional development. Educational media, such as educational television (TV) shows, interactive apps, and online resources, can offer valuable educational content [3]. It can foster creativity and imagination, as media about storytelling, animations, and imaginative play can further expand their creative horizons [4].

Nonetheless, these are concerns regarding the extent of time children spend on this platform and their potential exposure to inappropriate content including commercial influences [5]. The platform hosts billions of videos spanning various genres, including content that may be deemed unsuitable for kids, teenagers, and even adolescents. Interestingly, most of these types of videos are freely accessible with little or no effort. Inappropriate content on social media sites, such as YouTube, is an issue for parents, and children being exposed to inappropriate content is a big concern because they might see things like violence, bad language, or stuff that may harm their mental/physical growth. It is important to keep children away from such inappropriate stuff to help them stay happy and healthy [6]. Inappropriate content covers a broad spectrum of subjects, such as adult content, terrorism, racism, suicide, violence, and sexism, all of which can be harmful, especially for children [7–9]. Exposure to inappropriate content can negatively impact children's mental and physical development, leading to sleep disturbances, cyberbullying, abuse, and security risks. This is evident from the experiences of many teenagers who have come across such content, as reported in a study [10]. To tackle this issue, it is important to have systems that automatically check and control what people watch, especially for children, and to keep them safe and mentally/physically healthy [6]. To address the inherent risks associated with online content consumption, the *parental mediation theory* [11] has been introduced. This theory advocates parental mediation as an effective approach to mitigate the mentioned concerns. The theory is defined as "a strategy aimed at assisting children in utilizing and comprehending the internet, fostering positive outcomes, and preventing adverse effects of internet use" [11]. The parental mediation theory consists of three different approaches, each serving as a method for parents to navigate their children's digital experiences [12]. For instance, (a) active mediation: parents talk regularly as friends with their children about what they watch on their smartphones, tablets, and computers and play online. (b) Restrictive mediation: in this approach, the parents make strict rules about what their kids/children can and cannot watch or play online games. (c) Couse mediation: parents are involved and watch or play online content with their kids when they are using social media or the internet. It primarily involves nonverbal communication, as parents observe without necessarily engaging in active conversations about the content their children are watching [12].

Online mediation aims to mitigate the adverse impacts of information and communication technologies (ICTs) while harnessing their positive potential. A plethora of parental controls and mediation techniques have been introduced in pursuit of this goal. To develop an effective mediation strategy, it is essential to gain insight into the factors influencing parental decision-making in this context. In this context, the overarching aim of online mediation is to minimize the detrimental consequences associated with ICTs while optimizing their advantageous effects [12].

Numerous solutions in the form of software have been proposed to curtail access to undesirable content (see Tables 1, 2, and 3*)*. Nevertheless, a significant proportion of these solutions entail manual intervention, reliance on expensive hardware infrastructure, technical complexities, and time-intensive procedures and are hence rarely used. Consequently, there is a growing consensus in favor of automated methods for the detection and filtration of inappropriate content within video-sharing platforms. It is this compelling and pressing issue that serves as the driving force behind these research efforts as we try to augment content moderation and boost user safety in this digital age.

This research is aimed at filling notable grey areas within current solutions, which often fall short of comprehensively estimating users' ages and precisely classifying inappropriate video content. The motivation behind this approach is rooted in a commitment to safeguarding sensitive viewers and establishing a robust framework that leverages the capabilities of automated content analysis techniques. In alignment with the increasing emphasis on responsible content sharing, the objective is to contribute to a safer and more constructive online environment for all types of users, with a particular focus on children and adolescents.

In addition to the software solutions (discussed in Table 1), there are persistent challenges associated with children and teenagers accessing inappropriate content without constraints. We argue that existing solutions are, in certain instances, neither precise nor practical. Notably, these solutions are reliant on user familiarity with parental controls, making them less accessible to individuals lacking knowledge of such applications/software. Hence, there is a compelling need for the implementation of an automated model designed to detect and filter inappropriate content. This study advocates for the adoption of such an automatic content filtration system as a practical and effective means of addressing this issue. The main objectives of this study are as follows:

- To identify the shortcomings in existing deep learning solutions for detecting and classifying inappropriate videos.
- To design a deep learning–based framework, with a focus on age-based classification, for detecting and categorizing inappropriate videos.

This study looked to address a fundamental research inquiry: How can the integration of age-aware content filtering and advanced classification techniques be harmonized to develop a precise and user-friendly model for the detection and categorization of inappropriate content? The goal of this

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Table 1: Comparison chart of different parental control applications.

		Applications							
		Parental Control for Android	**Kaspersky Safe Kids**	**MMGuardian**	**Norton Family**	**OurPact**	**Qustodio**	**Screen Time**	**Net Nanny**
	Platform	**Android**	**Android, iOS, Mac, and Windows**	**Android and iOS**	**Android, iOS, and Windows**	**Android and iOS**	**Android, iOS, Kindle Fire, Windows, and Mac**	**Android and iOS**	**Android, iOS Kindle Fire, and Windows**
Features	Support for number of users	Unlimited	Unlimited	1–5	Unlimited	1–20	1–15	1–5	1–20
	Web portal for parents	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Call logging	None	None	Android	None	None	Android	None	None
	Text blocking	None	None	Android	Android	None	Android only	None	None
	Geofencing	Android only	Yes	None	None	Yes	Yes	Yes	None
	Location tracking	Android only	Yes	Yes	Yes	Yes	Yes	Yes Extra fee	Yes
	Web monitoring	Android only	Yes	Yes	Yes	None	Yes	Yes	Yes
	Time limits	Android only	Yes	Android only	Android only	Yes	Yes	Yes	Yes
	Scheduling	Android only	Yes	Yes (iOS only)	Yes	Yes	Yes	Yes	Yes

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Table 2: Parental controls in web browsers.

Features	Web browsers				
	Internet Explorer	Google Chrome	Firefox	Edge	Safari
Activity report	Yes	No	No	No	No
Web filtering time	Yes	Yes (with extension)	Yes (with extension)	Yes (with extension)	Yes (with extension)
Limits app	Yes	Yes (with extension)	Yes (with extension)	Yes (with extension)	Yes (with extension)
Management	Yes	No	No	No	No

Table 3: Two popular Chrome extensions.

Chrome extensions	Features				
	Website block by URL	Automatically block	Password protection	Grace period	Time limits
Block site	Yes	Yes	Yes	Yes	No
Stay focused	Yes	No	No	Yes	Yes

research attempt is to enhance online safety standards for children and teenagers.

In this context, the proposed research introduces an age-based deep learning approach designed to find inappropriate content. The proposed system is fully automated, requiring minimal manual intervention, and does not need technical expertise for its operation. This study represents a concerted effort to contribute to the field of parental mediation theory, offering a valuable tool for practitioners seeking to improve the media consumption habits of children and adolescents. To validate the proposed approach, we utilized a dataset generated from YouTube [13] comprising manually annotated videos, encompassing a diverse range of categories, including safe, real violence, drugs, nudity, simulated violence, kissing, pornography, and terrorism.

The proposed framework consists of two key modules: (a) user profiling and (b) detection/classification of inappropriate video content. In user profiling, the system employs a camera to capture user images, allowing age estimation through facial feature analysis. To achieve better accuracy, we utilized the specialized AgeNet model [14], which employs deep learning techniques and has been trained on annotated datasets, including the FG-NET[1] dataset. This training equips the AgeNet model to differentiate complex relationships between facial characteristics and estimated age. In detecting and classifying inappropriate video content, we used the latest state-of-the-art technologies to sort and organize these videos into different groups. To accomplish this objective, we harness the capabilities of two robust models: MobileNetV2 [15], which excels at efficient feature extraction from videos, and the Transformer model [16], known for detecting and categorizing content accurately. MobileNetV2 [15] is a neural network architecture that serves as the foundation for extracting valuable features from videos.

The proposed architecture is designed to maximize resource efficiency while ensuring better performance. By utilizing MobileNetV2, visual details can be effectively captured from the videos (frames). After extracting the video features, we utilized the Transformer model, known for its excellence in natural language processing tasks and its adaptability across various domains, including images and videos. This approach demonstrated impressive effectiveness in capturing complex patterns and relationships within sequential data. Using the Transformer model makes the proposed system significant by looking closely at video details (metadata) and detecting/classifying inappropriate content. The model understands the context and minor differences in video frames, helping it decide if they are appropriate or not. The proposed system does not only find inappropriate videos (content), but it also classifies them into eight different categories (mentioned above), which helps us understand them better. The significance of the proposed approach was demonstrated through the evaluation of diverse manually annotated datasets generated from YouTube. The model achieved an improved accuracy of 96.89% and an $F1$-score (5-fold) of 0.9034, confirming its high precision and reliability. It outperforms the existing solutions by comparing the results with the proposed system (see Results section).

The significance of the proposed framework is that it not only meets the need for accurate content classification but also contributes to the wider research gap in automated content moderation. It may serve as a protective shield for users, particularly children and adolescents, guarding against exposure to harmful and age-based inappropriate video content. In the future, the proposed framework can be used for real-time video streaming services for the seamless detection of inappropriate video content. This automation of parental controls may further enhance security and privacy and may overcome the negative impact of inappropriate video content.

The remainder of this paper is structured into five distinct sections. Section 2 is related to a detailed yet comprehensive "Literature Review," offering a thorough examination of the existing studies. Section 3 is related to "Proposed Methodology," providing an in-depth discussion of the proposed framework. Section 4 presents a detailed discussion of the "Results and Analysis" generated by the system, along with a comparative analysis against previous research studies. Section 5 is about the "Discussion" of the generated results. The last section is "Conclusion and Future Work," which concludes the

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paper. The References section includes all relevant sources cited throughout the paper.

2. Literature Review

The inappropriate content on the internet covers a broad spectrum of materials that are deemed inappropriate or potentially harmful for minors (kids), teenagers, and even senior citizens. These categories include content featuring explicit violence, offensive language, promotion of abuse, hate speech, and sexually explicit content [17].

A study [18] reported that excessive technology use and unrestricted internet access can negatively impact children's social, mental, and physical development. It may reduce social interactions, heighten aggression, lead to health problems, and disrupt sleep patterns [19]. Moreover, it exposes children to various risks, including cyberbullying, abuse, security threats, access to adult content, harassment, bullying, violence, and encounters with explicit sexual material, all of which are considered harmful and inappropriate for young audiences [20].

Sexual content not suitable for kids or minors refers to explicit or suggestive material made for adults [21]. This includes things like pornography or showing violence or sex that kids should not see. Seeing this kind of stuff in the early stages can affect kids' minds [22, 23]. Therefore, it is important to keep kids/minors away from such content.

Violence-related content in media, especially social media, can hurt viewers. These scenes, including intense violence, blood, injuries, and other disturbing images, can be upsetting even for younger viewers [24]. Parents, caregivers, content creators, and providers need to collaborate in implementing measures such as parental controls, age restrictions, and open dialogues to shield their children from the harmful effects of the mentioned violent content [25].

Similarly, hate speech refers to the use of language or expressions that provoke or advocate violence, bias, discrimination, or hostility directed at individuals or groups based on characteristics such as race, religion, civilization, gender, sexual orientation, or other defining attributes [26]. It involves spreading offensive and derogatory remarks, slurs, stereotypes, and inflammatory statements to belittle, degrade, or marginalize individuals or communities. Hate speech not only undermines social unity and raises division but also causes significant emotional and psychological harm to those targeted individuals [27]. Therefore, society needs to actively tackle and fight against hate speech through educational programs, awareness efforts, legal actions, and by promoting inclusivity, tolerance, and respect for diversity.

Moreover, the content related to drugs and alcohol includes materials that depict or promote the consumption of substances like illegal drugs and alcoholic beverages. These materials can manifest in various forms, including written content, visuals, videos, and promotional content. They may depict drug use, glorify substance abuse, or offer guidance on obtaining and using these drugs [28].

Inappropriate content also includes cyberbullying, which refers to the intentional act of harassing or intimidating others through digital platforms and online communication channels. It entails the utilization of technology, such as social media, messaging apps, or online forums, to single out individuals to inflict harm upon them [29]. Its repercussions can be profound, frequently resulting in emotional distress, anxiety, unhappiness, and tragically, even triggering suicidal thoughts among its victims [30]. Likewise, as digital technology advances and online platforms, such as social media, become more widely used, deceptive or misleading information can rapidly disseminate to a broad audience [31].

To fight against such issues, parental control applications have become essential tools for parents seeking to monitor and mitigate the potential adverse effects of online activities. The market offers a variety of mobile applications designed for parental control purposes. Tables 1, 2, and 3 provide a comparative analysis of different mobile phone parental control applications based on essential features. These features include a web portal for parental oversight, call and text message management, web monitoring, location tracking, time limits, and application management.

One noteworthy solution is "WebGuard," a filtering system precisely crafted to detect and filter inappropriate or adult content [32]. Additionally, another application showcases its ability to intercept and analyze images transmitted through various channels, including Bluetooth and Multimedia Messaging Services (MMSs) on mobile devices. This application not only blocks such images but also employs automated classification to identify and categorize them as inappropriate content [33].

Similarly, the traditional content filters embedded within operating systems (Windows, Linux, and MacOS) lack advanced capabilities. Parental control applications, while offering an additional layer of control, often consume significant system resources, prove ineffective, and can be bypassed swiftly. Moreover, these parental control solutions primarily operate on a local system level, which may not cover all online access points, especially considering that teenagers can access the internet from various devices and locations where parental control software may not be installed [34]. This gap leaves room for potential exposure to inappropriate content, and these parental control software options often come at a cost.

In response to these challenges, deep learning techniques have emerged as a promising approach in the field of video recognition for the automatic classification of inappropriate content. One approach utilizes a machine learning model that combines EfficientNet-B7 with bidirectional long short-term memory (BLSTM) to achieve precise identification and categorization [35]. In a recent advancement, the concept of "Private Parts Censor" (PP-Censor) emerged as an innovative solution. PP-Censor is a convolutional neural network (CNN)–based model meticulously designed for the real-time recognition and automatic censorship of adult content detected within video frames [36]. PP-Censor attempts to offer transparent and efficient solutions for the censorship of various types of inappropriate content.

Numerous smartphone-based applications for parental controls on Android and iOS also contribute to their market share. The parental control applications [37] comprise features designed to meet the viewer's requirements. Content filtering, web monitoring, app blocking, and screen time

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management stand as key components supporting numerous applications. It is essential to carefully consider factors such as social acceptability, user-friendliness, compatibility with devices, and exceptional customer support, which some of these applications lack.

In Table 2, the parameters, such as the "Activity Report" feature, involve the generation of a comprehensive report that offers insights into a user's online activities, including visited websites, time spent online, and accessed content. The "Web Filtering" feature refers to the ability to restrict access to specific websites or types of content based on predefined criteria, such as sex, nudity, and violence. The browsers, such as Google Chrome, Firefox, Edge, and Safari, all offer web-filtering capabilities, requiring the installation of extensions to activate this functionality. Similarly, the feature "Time Limits" refers to the capacity to set restrictions on the amount of time user(s), especially children and teenagers, can spend online. The "App Management" feature oversees and controls the usage of a specific application(s), including online games.

Google Chrome's parental control extensions (shown in Table 3) can also be used as a shield against harmful content, whether you are configuring a computer in a professional setting to restrict access to specific websites or you are a concerned parent aiming to ensure your child's safe online experience by keeping them away from inappropriate content [38]. The "Block Site" extension for Google Chrome offers users a comprehensive toolkit for efficient online activity management. This extension empowers users to proactively block specific websites by their URLs, effectively limiting access to undesirable or time-consuming content. Table 3 shows some of the features of Chrome's extensions.

On the other hand, the "Stay Focused" extension, although aligned in purpose, employs a distinct approach compared to the "Block Site" extension. While it does not provide direct website blocking through URLs, it introduces an innovative feature called the "Grace Period" to restrict inappropriate content.

TV parental guidelines serve as informative tools for parents, providing insights into the subject matter and appropriateness of TV shows for their children [39]. In conjunction with TV ratings, the V-chip [40], a standard component in most TVs, can be employed to empower parents to restrict access to content they consider unsuitable for their children [37]. These TV parental guidelines can also be seamlessly integrated with parental control technologies in cable TV and satellite set-top boxes, such as dish-TV, enabling the blocking/restricting of programs based on their respective ratings [41, 42].

Similarly, YouTube Kids [43] represents a child-centric online platform designed to offer a secure and regulated online space for children to explore age-appropriate video content. To keep a safe space for kids, the platform uses both automated systems and human reviewers to choose and filter videos, making sure they are right for children. This helps prevent them from seeing anything inappropriate or harmful, making their browsing experience safer. However, it is important to note that the platform includes ads, including sponsored content, which sometimes can interrupt what kids are watching and might expose them to inappropriate ads [44], such as drug promotional ads.

3. Proposed Methodology

In the proposed framework, shown in Figure 1, two fundamental modules are used: (1) user profiling and (2) detection and classification of inappropriate video content. In user profiling, the proposed system uses the capabilities of a camera to capture the user/viewer's image and estimate their age by examining facial features. The pursuit of precise age estimation leads us to embrace the specialized AgeNet model [14], which is an example of a deep learning model. AgeNet was trained on annotated datasets, among which the venerable FG-NET dataset takes its place, providing the base for understanding the complicated interplay between facial characteristics and the age of the viewer. The study uses a hybrid approach by harnessing the capabilities of two powerful models: MobileNetV2 and the Transformer model. MobileNetV2 [15], a state-of-the-art neural network architecture, leads to the efficient extraction of features from videos.

The main focus during implementation was on resource efficiency, which the proposed system maintains with significant performance while thoroughly analyzing video content. Leveraging the Transformer model, renowned for natural language processing and adaptability in various domains, it excels at uncovering complex patterns and relationships within sequential data. At the front, the Transformer model enables the system to conduct in-depth video feature analysis, ensuring accurate identification of inappropriate content. Its contextual understanding and sensitivity to understated differences establish it as a reliable evaluator of video appropriateness.

Beyond detection, our system takes a step further by categorizing inappropriate videos into eight distinct categories. This refined classification enhances our understanding of content, facilitating informed decisions and actions. Through the seamless integration of MobileNetV2's feature extraction capabilities and the analytical finesse of the Transformer model, our proposed framework serves as a guardian, promoting online safety and enabling effective content moderation. Through the seamless integration of these components, our proposed framework ensures reliable detection of inappropriate content, playing a pivotal role in enhancing online safety and refining content moderation practices. Additionally, in the following sections, we provide detailed explanations of each aspect, ensuring a thorough understanding of the methodology. This approach empowers readers to gain deep insights into the precise procedures involved, promoting clarity and facilitating the reproducibility of our research and results. By offering comprehensive descriptions, we aim to promote transparency and provide a comprehensive guide for the seamless implementation of our proposed methodology, as shown in Figure 1.

3.1. Face Detection. In this phase, the proposed methodology utilizes the functionality of a camera seamlessly incorporated into a mobile device (smart TV, laptop, and smartphone) to capture a high-quality image of the viewer. This captured

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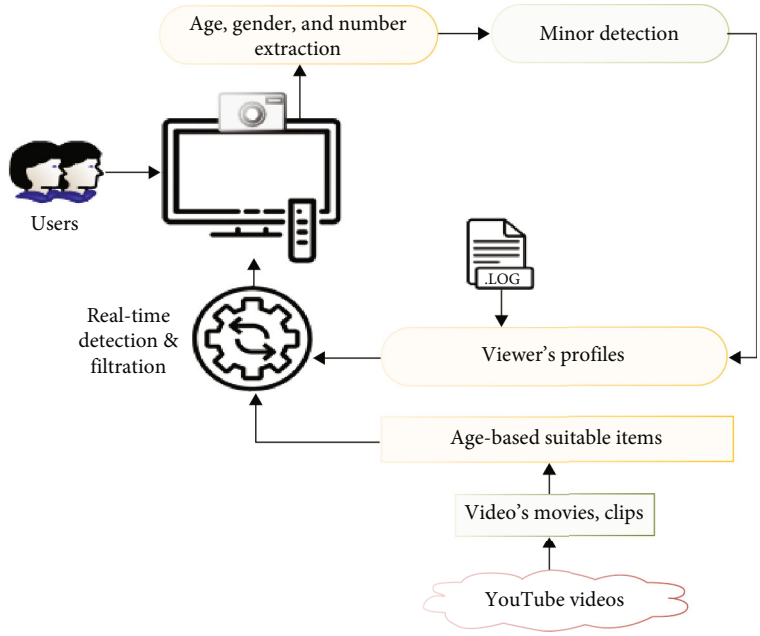


Figure 1: Proposed architecture and working mechanism.

image acts as the principal input for the subsequent age estimation process, skillfully performed by the proficient AgeNet model.

To meet the exact specifications of the AgeNet model, the captured image undergoes thorough preprocessing procedures. The process of image preprocessing encompasses a sequence of designed transformations to improve image clarity and extract prominent facial characteristics. Initially, the captured image undergoes adjustments to accommodate variations in lighting conditions, ensuring its integration into a uniform format. This normalization procedure effectively mitigates the influence of changing lighting conditions on subsequent processing.

Subsequently, the captured image undergoes facial detection and alignment processes, precisely identifying and aligning the facial region. This precise alignment eliminates potential distortions caused by varying head poses, resulting in more accurate age estimation outcomes. With the facial region well isolated and aligned, subsequent preprocessing steps prepare the image for seamless integration with the AgeNet model. Typically, this optimization involves resizing the image to meet the model's specific resolution requirements, ensuring efficient processing while preserving critical facial details. The thorough image preprocessing underscores our methodology for preparing the image to maximize the AgeNet model's effectiveness. This careful attention to detail in image preprocessing plays a crucial role in ensuring dependable and accurate age estimations, thereby enhancing the overall effectiveness of our user-profiling system, which further enhances the system's capabilities.

3.2. Age Estimation. AgeNet, our core module, is a specialized neural network selected with great care for its exceptional ability in accurate age estimation. This model exemplifies precision and is accurately crafted and trained to analyze facial features and accurately determine a person's age [45]. AgeNet's training encompasses annotated datasets such as the distinguished FG-NET dataset, refining its ability to provide reliable and precise age predictions. Its architecture features a combination of convolutional layers, pooling layers, and fully connected layers, working in harmony to extract subtle features and make age predictions based on input facial images. The depth of AgeNet's deep learning capabilities, a key aspect of its design, enhances our methodology for accurate age estimation, thereby enhancing the effectiveness of our user profiling system.

3.3. Minor Detection. In addition to age estimation, the proposed methodology encompasses a critical step that focuses on whether the user/viewer is a child or an adult, leveraging insights from their facial attributes. This aspect is pivotal for delivering a more personalized user experience and tailored content suitable for each age group. To accomplish this task, the system utilizes the robust capabilities of our trained model, which is accurately designed to differentiate between a child and an adult.

The training of the proposed model involves a diverse range of datasets, each precisely annotated with examples of both child and adult faces. Through exposure to the dataset, the model develops a keen sensitivity to the understated features and patterns unique to each age group. When it comes to detection, the model analyzes the facial image captured by the camera, employing advanced algorithms to identify key facial characteristics that distinguish children from adults(s). These distinguishing features may include variations in facial proportions, specific facial attributes, or other visual cues that distinctly reflect age-related disparities.

3.4. Inappropriate Detection and Classification of Videos. In the proposed methodology, this step performs the crucial

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task of identifying and categorizing inappropriate video content, a task essential for ensuring online safety. This complex process relies on cutting-edge technology to pinpoint and sort video appropriately. Once the system extracts the video features, the Transformer model performs its task. Additionally, it involves categorizing inappropriate videos accurately. This categorization, crucial to our approach, offers a detailed understanding of the content, aiding in subsequent actions and decisions. By comprehensively addressing different forms of inappropriate content, our methodology strengthens content moderation efforts, ensuring online spaces remain safe and inclusive.

3.5. YouTube Video Sharing Platform. A notable contribution to the research community is the YouTube-8M[2] benchmark dataset, developed by Google. This huge dataset comprises over 8 million video IDs, each accurately categorized across a wide spectrum of classes. Serving as the basis for many studies, it offers researchers a broad and diverse platform for exploration.

Furthermore, alongside the YouTube-8M dataset, there exist other benchmark datasets tailored to specific niches within the video domain. These datasets cater to specialized domains, focusing on categories such as sports, action recognition, face recognition, sentiment analysis, video captioning, and more. Each of these datasets, whether it is Sports-1M [46], UCF-101 [47], HMDB51 [48], Kinetics [49], Face Recognition YTF [50], YouTube Celebrities [51], or MSVD [52] and MSR-VTT [53] for captioning, plays a pivotal role in advancing research within its designated domain. These datasets collectively provide a diverse array of opportunities, enabling researchers to meticulously analyze and comprehend the multifaceted realm of online videos. It is noteworthy that existing benchmarks and datasets do not specifically address child-inappropriate content.

3.6. Dataset Creation. To tackle the challenge of identifying disturbing content on YouTube, we have curated an accurately annotated video dataset. This dataset serves as a comprehensive repository of videos containing disturbing or inappropriate content. This dataset serves as a crucial asset in our ongoing efforts to cultivate a safer and more child-friendly environment on YouTube.

The dataset was created by sourcing data from YouTube (a video-sharing platform). For data collection, we utilized the framework and application programming interfaces (APIs) mentioned in the study [13]. Comprising a collection of 40,000 videos, the dataset is thoughtfully organized into eight distinct categories, as shown in Table 4. Each video in the dataset has been carefully assigned to one of these predefined categories to accurately depict its content. The dataset utilized in this study has undergone a thorough and labor-intensive curation process. It began with manually downloading videos from the YouTube platform. Subsequently, each video was accurately segmented into discrete 1-s segments. These segmented clips then underwent a comprehensive manual annotation process to accurately classify them into predefined categories. The dataset includes the following categories:

Table 4: Dataset distribution.

Classes	Assigned labels	Number of clips
Kissing	0	5000
Horror	1	5000
Nudity	2	5000
Drugs	3	5000
Real violence	4	5000
Simulated violence	5	5000
Safe	6	5000
Terrorism	7	5000
Total clips		40,000

The video selection was done based on specific criteria aligned with the research objectives, emphasizing factors such as relevance, content diversity, and the potential representation of predefined categories. To ensure uniformity and consistency across the dataset, all videos were standardized to a 1-s duration. This standardization streamlines the subsequent handling and processing of videos during the analysis and model development phases. Additionally, frames were extracted from each 1-s video at a frame rate of 25 frames per second (fps), resulting in a comprehensive collection of frames that adeptly encapsulate the visual content of the videos.

3.7. Feature Extraction. To derive meaningful video classification features, the system determines the stopping layer for feature extraction, opting for "block-13-expand-relu" as it captures high-level representations critical for the preceding tasks. This layer choice ensures the retention of rich visual information. The "feature extractor" [54] model is designed by truncating the original [55] at the chosen stopping layer. This model preserves layers up to and including the stopping point while discarding the classification-related layers. Operating on (224, 224, 3) input images, the feature extractor generates relevant feature maps for video classification. To maintain the integrity of pretrained weights, the system freezes the layers of the MobileNetV2 model, preventing gradient updates during backpropagation [55]. This not only safeguards learned representations but also enhances computational efficiency during subsequent video classification model training. Figure 2 shows some pretrained model comparisons.

4. Results and Analysis

We employ an AgeNet model, trained on the UTKFace[3] dataset, to predict user age and gender, providing crucial context for subsequent video content analysis. The proposed video classification component relies on a manually curated dataset of 40,000 YouTube videos, spanning eight categories, each restricted to 1-s duration. For feature extraction, we turn to the MobileNetV2 model, renowned for its ability in this domain. Extracted video features are saved as NumPy arrays and serve as inputs for the proposed bespoke Video-Transformer model.

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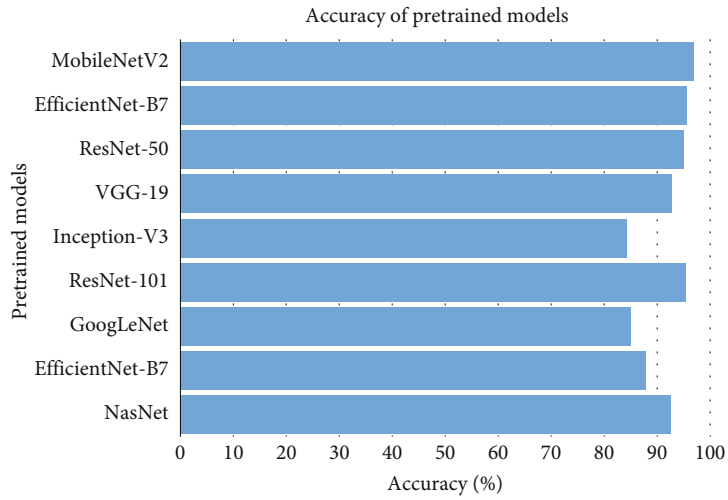


Figure 2: Pretrained model comparison.

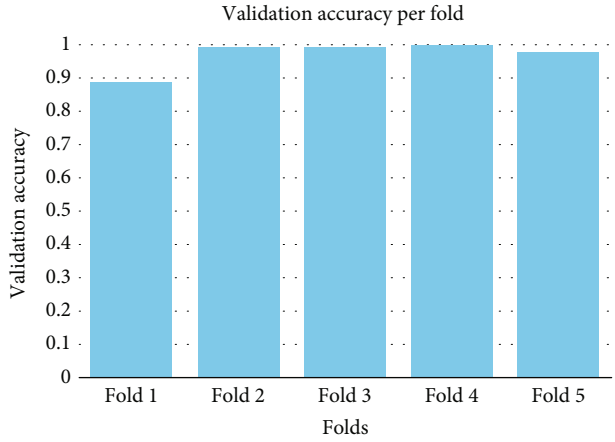


Figure 3: *K*-Fold Cross-Validation.

The forthcoming sections provide a detailed analysis of each stage of the model, accompanied by a comprehensive evaluation of the results. We initiate the process with an assessment of face detection and age-gender prediction performance, as conducted in the study [56]. This journey concludes with a thorough evaluation of the video classification capabilities of the proposed custom VideoTransformer model, resulting in commendable outcomes and performance.

4.1. Data Collection and Preprocessing. To construct a diverse and representative dataset, we collected videos from YouTube across eight distinct categories: *kissing*, *real violence*, *nudity*, *simulated violence*, *drugs*, *terrorism safe*, and *horror*. These categories were carefully chosen to encompass a wide spectrum of potentially inappropriate online content. Throughout the dataset collection process, we adhered rigorously to ethical and legal guidelines, ensuring compliance with community standards, copyright laws, and privacy regulations. Priority was given to safeguarding individuals' privacy and identity.

All videos underwent resizing to a standardized dimension of 224×224 pixels, featuring three color channels ($224 \times 224 \times 3$). This standardization ensures consistency across the dataset, facilitating efficient feature extraction. To maintain uniform temporal context, each video was adjusted to a fixed duration of 1 s. This uniformization ensures that all videos share equivalent time intervals for subsequent analysis and comparison. Normalization techniques were applied to the video frames to enhance data consistency, involving the transformation of pixel values into a common color space and the normalization of intensity ranges.

During the data preprocessing phase, stringent quality control measures were implemented. These measures included manual validation of category labels, elimination of duplicates or irrelevant videos, and mitigation of any potential dataset biases. These steps were critical in maintaining the integrity and reliability of the collected dataset.

4.2. K-Fold Cross-Validation Results. Integrating *K*-fold cross-validation into the code improves the model's robustness by dividing the dataset into five folds, with each fold used for validation in a separate iteration. The following bullets and Figure 3 outline the validation accuracy for each fold, providing insight into the model's performance across different subsets of data:

- Fold 1: achieves a validation accuracy of 0.88916, indicating how well the model generalizes to this particular subset.
- Fold 2: impresses with a validation accuracy of 0.994141, signifying high accuracy in predicting the target variable for this fold.
- Fold 3: maintains strong generalization capabilities with a validation accuracy of 0.992188 on this subset.
- Fold 4: excels with an exceptional validation accuracy of 0.99967, demonstrating the model's prowess on this data.

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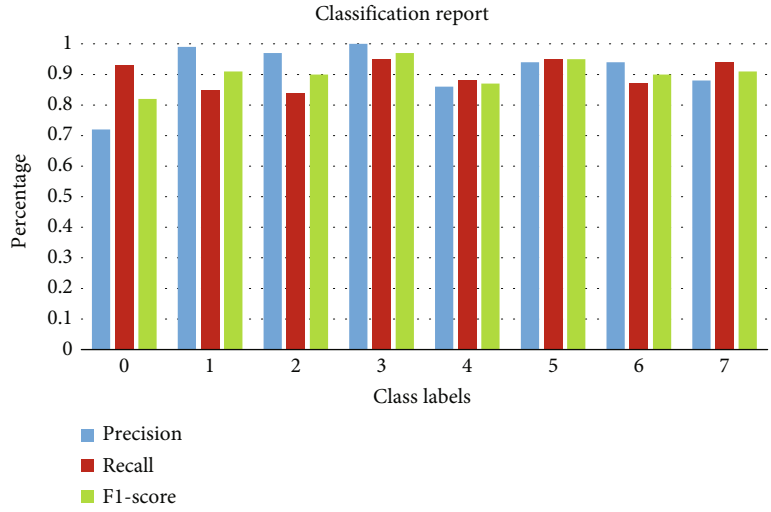


Figure 4: Precision, recall, and accuracy.

- Fold 5: shows superior performance with a validation accuracy of 0.978271 on this subset.

4.3. Classification Report. The classification report offers a comprehensive evaluation of the classification model's effectiveness across multiple classes. It includes overall metrics for the entire dataset as well as specific metrics for each class, such as precision, recall, and $F1$-score. Figure 4 illustrates performance metrics (precision, recall, and $F1$-score) for various class labels, assessing the model's accuracy in both identifying positive instances and capturing actual positive instances. The $F1$-score, serving as a balanced performance measure, harmonizes precision and recall to provide a comprehensive evaluation.

4.3.1. Precision [57]. Equation (1) stands as a pivotal performance metric for classification models, evaluating their accuracy in predicting positive cases. It measures the ratio of true positive predictions to the sum of true positives and false positives. The higher precision values signify fewer incorrect positive predictions. Mathematically, the precision metric is calculated as follows:

$$\text{Precision} = \frac{\text{true positives}}{(\text{true positives} + \text{false positives})}. \quad (1)$$

4.3.2. Recall [57]. Equation (2), referred to as sensitivity or true positive rate, evaluates a model's capability to accurately identify positive instances within the actual data. It is computed by dividing the sum of true positives and false negatives by the total number of true positives. A higher recall score indicates fewer false negatives, signifying that the model captures a larger proportion of actual positive cases. Mathematically, recall is expressed as follows:

$$\text{Recall} = \frac{\text{true positives}}{(\text{true positives} + \text{false negatives})}. \quad (2)$$

4.3.3. F1-Score. Equation (3) evaluation metric [57] is a metric that balances a model's recall and precision, which is particularly useful when both false positives and false negatives are significant. It is the harmonic mean of precision and recall and provides a single score that reflects the model's overall performance in classification tasks. The $F1$-score reaches its best value at 1 and worst at 0, whereas the higher values indicate better model performance. Mathematically, the $F1$-score is calculated as follows:

$$F1\text{-score} = 2 * \left(\frac{(\text{precision} * \text{recall})}{(\text{precision} + \text{recall})}\right) \quad (3)$$

In Figure 4, the model's performance across different class labels is as follows:

- For Class Label 0: precision = 0.72, recall = 0.93, $F1$-score = 0.82
- For Class Label 1: precision = 0.99, recall = 0.85, $F1$-score = 0.91
- For Class Label 2: precision = 0.97, recall = 0.84, $F1$-score = 0.9
- For Class Label 3: precision = 1.0, recall = 0.95, $F1$-score = 0.97
- For Class Label 4: precision = 0.86, recall = 0.88, $F1$-score = 0.87
- For Class Label 5: precision = 0.94, recall = 0.95, $F1$-score = 0.95
- For Class Label 6: precision = 0.94, recall = 0.87, $F1$-score = 0.9
- For Class Label 7: precision = 0.88, recall = 0.94, $F1$-score = 0.91

In summary, the model's performance varies across class labels. Some classes, such as Labels 1, 3, 5, and 7, exhibit

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Table 5: Performance comparison between the proposed MobileNetV2-VideoTransformer model with existing state-of-the-art video classification techniques.

Studies/approaches	Pretrained model	Modality	Epochs	Classes	Accuracy (%)	Precision (%)	Recall (%)	*F*1 -score
[35]	NasNet	Video	5-fold	2	92.6	—	—	0.88
[47]	EfficientNet-B7	Video	5-fold	2	87.87	86.65	89.12	0.878
[47]	GoogLeNet	Video	200	2	85.1	—	—	0.882
[50]	ResNet-101	Video	5-fold	2	95.5	—	—	—
[58]	Inception-V3	Multimodal	—	2	84.3	82.1	89.0	0.829
[59]	VGG-19	Multimodal	25	4	92.83	92.72	91.71	—
[48]	ResNet-50	Multimodal	10	2	95	—	—	—
BLSTM	EfficientNet-B7	Video	20	3	95.66	93.17	92.22	0.9267
Proposed VideoTransformer = 128	**MobileNetV2**	**Video**	**20**	**8**	**90**	**—**	**—**	**0.9034**
Proposed VideoTransformer = 128	**MobileNetV2**	**Video**	**5-fold**	**8**	**96.89**	**92**	**91**	**0.9034**

Note: The last row (in bold font) depicts the proposed video transformer with 5-fold, achieving an accuracy of approximately 96.89%. This accuracy demonstrates significance compared to other state-of-the-art studies listed in the table.

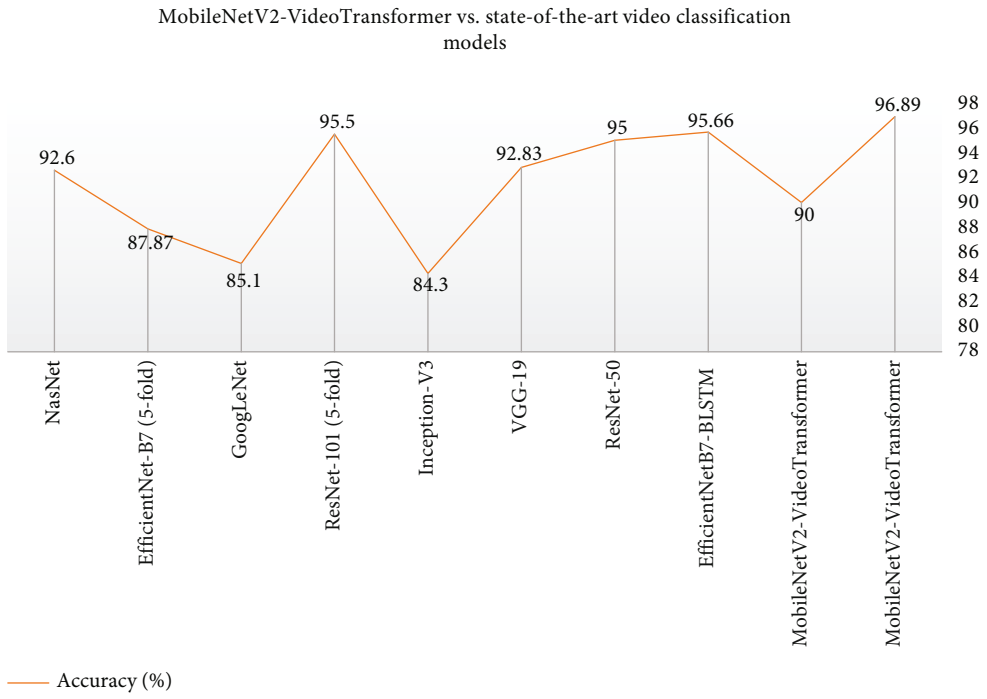


Figure 5: Proposed work VS state-of-the-art results.

exceptional precision, recall, and *F*1-scores, indicating accurate identification, with scores ranging from 0.88 to 0.99 for precision, 0.85 to 0.95 for recall, and 0.91 to 0.97 for *F*1 -scores. Conversely, Classes 0, 2, 4, and 6 show slightly lower performance metrics, with precision ranging from 0.72 to 0.94, recall from 0.84 to 0.93, and *F*1-scores from 0.82 to 0.9. Despite these variations, the model demonstrates overall strong performance, with high precision, recall, and *F*1 -scores for most class labels, making it suitable for real-world video classification applications.

5. Discussion

To assess the effectiveness of the proposed MobileNetV2-VideoTransformer model in classifying inappropriate content, a comparative analysis is conducted with several state-of-the-art models/studies and techniques commonly used in similar tasks. Table 5 summarizes the results and performance ratings of different existing and proposed classification methods. It is worth mentioning here that prior research has explored diverse modalities of YouTube data, including text, audio,

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video, and metadata, for content classification. Among these studies, a prevalent approach for identifying unsafe content involves employing pretrained CNN models with either machine learning–based classifiers or LSTM–based classifiers like support vector machines (SVMs) [60–62].

Unlike methods that rely on pretrained CNN features coupled with SVM classifiers, the proposed MobileNetV2-VideoTransformer approach achieved significantly higher accuracy compared to GoogLeNet-SVM [61], fine-tuned NASNet-SVM [62], and EfficientNet-SVM [60] methods, demonstrating significant improvements of 3.06%, 7.79%, and 10.1%, respectively. When compared to baseline models incorporating pretrained CNNs and LSTM neural networks, the Inception-V3 with an LSTM–based approach [63] achieved a lower $F1$-score of 0.828. Conversely, the proposed MobileNetV2-VideoTransformer model showed superior performance in the realm of video classification. This superiority was evident both with its attention-based variant ($F1$-score = 0.9195) and without its attention-based variant ($F1$-score = 0.9267).

It is important to acknowledge here that previous studies, such as those employing the ResNet-LSTM model [63, 64], achieved comparable accuracy results to the proposed technique. This similarity can be attributed to the fact that these studies primarily focused on binary video classification, which is simpler compared to multiclass video classification. However, the proposed MobileNetV2-VideoTransformer model surpasses these approaches, indicating the robustness of BiLSTM in modeling time-series data, especially in multiclass video classification tasks, as opposed to VGG-LSTM [43, 65]. Furthermore, other studies that utilized simple CNNs [66, 67] reported lower classification accuracy and $F1$-scores compared to alternative methods. This highlights the inadequacy of straightforward CNN models in capturing the complexities inherent in YouTube data. Figure 5 shows the comparison of proposed models with state-of-the-art studies.

In conclusion, the proposed MobileNetV2-VideoTransformer model, which combines efficient feature extraction from MobileNetV2 with attention-based sequence modeling from the Transformer, outperforms previous studies (shown in Figure 5) in the detection and classification of child-inappropriate video content. The performance comparison results confirm the efficacy of the proposed model in this specific domain.

The proposed model exhibits several notable strengths that contribute to its effectiveness in detecting inappropriate video content. These strengths include high classification accuracy, resilience against common forms of inappropriate content, scalability to handle extensive datasets, adaptability to diverse modalities and domains, and the ability for continuous learning and improvement through iterative training. These attributes position the proposed model as a valuable tool for content moderation, user safety, and the promotion of a safer online environment.

A closer examination of the classification report reveals variations in precision, recall, and $F1$-scores across different categories. Categories 1, 3, 5, and 7 demonstrate superior performance in classifying inappropriate content, while Categories 0, 2, 4, and 6 exhibit lower scores, indicating room for refinement. Throughout the cross-validation process, the model consistently demonstrates high accuracy across the folds, with Fold 2 achieving the highest accuracy at 99.41%, while other folds maintain accuracies above 97%.

It is essential to mention the limitations of the proposed model in detecting inappropriate video content. These limitations include potential variations in performance across languages and cultures, the challenge of identifying nuanced content, difficulty in detecting emerging forms of inappropriate content, and susceptibility to adversarial attacks. Recognizing and addressing these limitations is crucial for ongoing research and improvement to ensure the model's efficacy in real-world scenarios. Inappropriate video content can exhibit variations in visual content, context, and explicitness, introducing inherent noise and ambiguity that pose challenges for accurate classification. Enhancing the model's ability to handle noise and recognize subtle cues or contextual clues can further improve its performance.

Moreover, it is worth noting that training and deploying video classification models can be computationally demanding, requiring substantial hardware resources and time. Scaling the model to accommodate larger datasets or real-time video analysis may encounter computational constraints.

6. Conclusion and Future Work

The research has concluded in the development of a robust model for the detection and classification of inappropriate video content, showing competitive performance in accuracy. However, it encounters challenges associated with data imbalance, noise, and computational resource requirements. This study addresses the pressing need to enhance content moderation and user safety on video-sharing platforms, particularly concerning the vulnerable demographics of children and even adults. In an era characterized by the rapid expansion of digital media and social networking sites, where young users are susceptible to exposure to potentially harmful/inappropriate including age-inappropriate content, this research provides a robust solution leveraging advanced technologies to safeguard online experiences. Through a two-module (hybrid) methodology, this study introduces a state-of-the-art framework that advances user profiling through age estimation and facilitates the accurate detection and classification of inappropriate video content. The effectiveness of the proposed approach was rigorously demonstrated through evaluations using manually annotated datasets from YouTube, achieving an impressive accuracy of 96.89% and an $F1$-score of 0.9034, underscoring its precision and reliability. Importantly, the proposed framework not only fulfills the need for accurate content classification but also addresses the broader research gap in automated content moderation. By automating content analysis, the proposed approach offers an efficient, scalable solution to enhance the safety of video-sharing platforms, especially for children and adolescents, thereby promoting responsible digital content consumption. As digital platforms, especially social media, continue to evolve, this research may lay the base for cultivating a safer online environment. Future research may need to refine the framework, adapt to

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emerging content types, and explore the seamless integration of richer user profiling.

Future research can be carried out to investigate advanced deep learning architectures, incorporate natural language processing techniques, integrate user feedback, and conduct comprehensive evaluations of various detection models. Incorporating user feedback mechanisms has the potential to enhance performance and may decrease false positives or negatives. Furthermore, integrating explainability techniques can illuminate the model's decision-making process, enhancing transparency and user trust. Moreover, the exploration of multimodal approaches that combine visual and audio analysis can significantly enhance the accuracy and reliability of inappropriate video content detection. Furthermore, the fusion of visual cues with audio features, speech recognition, and natural language processing can yield more comprehensive results. Efforts should also focus on developing scalable architectures and efficient algorithms capable of managing large-scale video datasets and enabling real-time processing. This may involve a survey of distributed computing frameworks, parallel processing techniques, and optimized deep learning architectures.

Data Availability Statement

The dataset is available on a public repository, namely, Kaggle, and can be accessed at the following link: "https://www.kaggle.com/datasets/15bbced86e095b0d05b9fbd554427061ed7f66e1402cb19d3ab8ddcc50b95c81."

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

All the authors listed in the manuscript have made significant scientific contributions to this research. Here is a summary of their specific contributions. *Iftikhar Alam* contributed to the subject area, helped in writing the manuscript, contributed to the paper's structure, and played a role in the results and analysis of the research. *Abdul Basit* conducted a comprehensive literature survey related to the research, summarized the relevant literature, and worked on machine learning training and testing using Python code. *Riaz Ahmad Ziar* was involved in dataset preparation, handled data preprocessing and cleaning, which was critical for the data-driven research of this study, contributed to the optimization process, and engaged in manual annotation within the dataset.

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Endnotes

[1]https://paperswithcode.com/dataset/fg-net
[2]https://paperswithcode.com/dataset/youtube-8m
[3]https://paperswithcode.com/dataset/utkface

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